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// Ruifeng Song 06/02/2025
#include "Particle.h"

// Constants
const size_t read_buf_size = 300; // Increased buffer size to accommodate
large chunks of data

// Buffer and index for incoming data from Serial1
char readBuf[read_buf_size];
size_t read_buf_index = 0;

SYSTEM_MODE(AUTOMATIC); // Automatic management of cloud connections

void setup() {
  Serial1.begin(9600);  // Start serial communication on Serial1 at 9600
baud
  Particle.publish("Setup", "Setup Done", PRIVATE); // Publish a test
event with test data
  Serial1.flush();
  Serial1.print("$");   // tell Arduino ready to receive data
}

void loop() {
  receiveBuffer();      // Handle incoming data from Serial1
}

void receiveBuffer() {
  while (Serial1.available() > 0) {  // Check if data is available on
Serial1
    char c = Serial1.read();          // Read a character from Serial1

    if (read_buf_index < read_buf_size - 1) {
      if (c == ';') {                 // Check for semicolon indicating
the end of the data
        readBuf[read_buf_index] = '\0'; // Null-terminate the string
        Particle.publish("Environmentdata", readBuf, PRIVATE); // Publish
the complete message
        delay(5000);
        read_buf_index = 0;           // Reset buffer index for the next
sample
        Serial1.flush();
        Serial1.print("#");           // tell Arduino ready to turn off
      } else {
        readBuf[read_buf_index++] = c; // Add character to readBuf
      }
    } else {
      Particle.publish("Error", "readBuf overflow, emptying buffer",
PRIVATE); // Publish error, buffer overflow
      read_buf_index = 0;             // Reset index if overflow occurs
    }
  }
}

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